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# Nick Petalas

Senior Software Engineer

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## EMPLOYMENT

**Senior Frontend Developer** | Eeze (Game Innovation) — Sep 2025 – Present

- Lead frontend developer across multiple concurrent iGaming projects, building with **React**, **Zustand**, **Protobuf**, and **Docker**.
- Led the end-to-end frontend development of **Operation Tools**, coordinating closely with backend engineers for seamless integration.
- Sole developer and project lead on the **Backoffice Template Service** — designed the architecture, implemented localization, comprehensive test coverage, and a multi-stage Docker deployment pipeline. The service became a reusable platform adopted across multiple products.
- Co-delivered the **MMD** and **M12D** iGaming projects, integrating template service support and contributing through to launch.
- Selected to lead frontend development on an upcoming iGaming project.

**Founder & Lead Engineer** | DailyGoal.fit — Feb 2025 – Present

- Founded and shipped a **health and fitness tracking platform** across web and **iOS** (live on the App Store), focused on simplicity, consistency, and meaningful user outcomes.
- Defined product vision and oversaw all aspects of engineering, UX design, and long-term roadmap planning.
- Designed and implemented scalable full-stack architecture using **Next.js**, **React Native (Expo)**, **Convex**, and **Vercel**.
- Built and shipped an iOS app with nutrition tracking, barcode scanning, workout logging, custom goals, Apple Health integration, streak systems, and in-app subscriptions.
- Built core features including food logging, goal tracking, workout tracking, and barcode scanning with integrations like USDA FoodData Central and OpenFoodFacts.
- Designed real-time database schema to support efficient querying, user-specific goal logic, streak tracking, and historical analytics.
- Developed **CI/CD workflows** and infrastructure automation including automated TestFlight builds and deployments.
- Led user research, customer support, and iterative product improvements based on feedback.
- Drove growth strategy, marketing initiatives, and early-stage community building.

**Senior Software Engineer** | Genesis Global Technology LTD — Dec 2019 – Oct 2024

- **Technical Lead** for the "Learn" team and core member of the web platform team over a 5-year tenure.
- Led the modernization of the web platform across three generations — migrating from legacy **Angular** to **StencilJS**, then to **Microsoft FAST** with custom web components and a bespoke design system.
- Collaborated with client teams on **React** integrations, bridging platform and product engineering.
- Mentored developers, onboarded new starters, and served as tier 3 escalation point for platform issues.
- Built developer tooling including **Genx CLI**, low-code scaffolding, app seeds, and custom **Docusaurus** plugins.
- Owned DevOps responsibilities including **AWS Amplify**, **GitHub Actions** pipelines, and Infrastructure as Code.

**Software Engineer** | Elemental Concept LTD — Sep 2018 – Dec 2019

- Full-stack engineer delivering across mobile, web, and backend on four concurrent client projects.
- Led major refactoring and microservice extraction on **Gift & Go**, a B2B rewards platform (**Java 11**, **Spring Boot**, **Angular**, **AWS**, **Docker**) — built APIs for enterprise clients including **Scientific Games**.
- Shipped **CorResilience** (health coaching app with lab and Fitbit integration) and **WellteQ** (wellness platform — Ionic, Cordova, Node.js, Java, MySQL, MongoDB).
- Built **Go Fast**, an internal rapid prototyping toolset (React, Angular, NestJS).

**Software Engineer** | Enigma Software Solutions LTD — Feb 2017 – Sep 2018

- Built and maintained iGaming platform tools for online betting clients: client portals, payment systems, trader and bet management, reporting, and retail cashier systems.
- Focused on **Java** and **SQL** back-end development.
- Maintained legacy systems (PrimeFaces, Oracle WebLogic).

**Software Engineer** | Safemarket LTD — Dec 2014 – Oct 2016

- Designed and developed internal tools and knowledge extraction systems for the Erybo project (**Java**, SQL, RDF, Jsoup, Selenium, Apache Jena).
- Built automated media processing tools (Java, **ffmpeg**), trivia content generators, and entity parsers.
- Developed services and parts of the Quizedia Android app.
- Trained and supervised interns in scraping and quiz generation.

## EDUCATION

**University of Macedonia** | Thessaloniki, Greece — Oct 2012 – Oct 2015

BSc In Applied Informatics. Final Grade: 7.7 / 10
Thesis: A study on Optimization of Object-Oriented Design using Genetic Algorithms.
Grade: 10 / 10; Supervisor: Alexander Chatzigeorgiou.

## OTHER TECHNICAL EXPERIENCE

I'm proficient across Windows, macOS, and Linux, and prefer working in the command line whenever possible. I have a strong personal interest in systems architecture, networking, filesystems, databases, and automation. I'm continuously learning and exploring new technologies through hands-on personal projects.

**Selected Projects:**

- **Homelab:** Built and maintained a Proxmox-based homelab running multiple VMs, including:
  - **TrueNAS VM** with HBA passthrough for an 8-drive **RAIDZ2 NAS**.
  - **Debian VM** running dockerized services: Traefik, Plex, Radarr, Sonarr, Lidarr, SABnzbd, qBittorrent, Filebrowser, Glances, Netdata, Home Assistant, and more.
  - Managed my own network stack: OPNsense router (on N100 box), Pi-hole for DNS-level adblocking, Unbound as recursive resolver, and Omada SDN Controller for access point management.
- **Heuristic Vector Image Generator:** Created a genetic algorithm to recreate images using layered semi-transparent polygons. Developed in Java (multithreaded), with an in-progress **Rust rewrite using portable SIMD**. A web-based version is available at artgen.xyz.
- **FreeCell Solitaire Solver:** Wrote a solver for FreeCell puzzles using custom heuristics and graph traversal algorithms (BFS, DFS, A*, Best-First). (Java)
- **Banknote Authentication:** Designed a neural network in MATLAB to classify banknotes as genuine or forged, achieving 100% accuracy on a UCI dataset.
- **Custom Database Merger:** Built a system to augment a large RDF database (based on Freebase) by scraping websites, performing entity recognition (OpenNLP), and fact extrapolation. (Java, MySQL, RDF)
- **BTC Risk Metric:** Developed a custom Bitcoin risk oscillator, combining metrics like EMA ratios, RSI, Mayer Multiple, Puell Multiple, and others to assess cyclical risk. Published as a TradingView indicator.
- **OpenSCAD Modeling:** Created programmatic 3D models for printing using OpenSCAD, combining scripting with parametric design.
- **Auto Conf Scripts:** Maintained a cross-platform setup automation suite for Linux/macOS. It installs hundreds of dependencies, patches fonts, configures zsh, clones repos, and links dotfiles to streamline new machine provisioning.

## PROGRAMMING LANGUAGES

- **Primary focus:** TypeScript / JavaScript (React, React Native, Node.js, and experience with multiple other frameworks and native web components), HTML, CSS, Bash (scripting), Dockerfiles.
- **Previously used professionally:** Java, Kotlin, Python, SQL.
- **Limited experience:** C, C++, R, PHP, MATLAB, SPARQL.
- **Currently learning:** Rust and Elixir.
- Always eager to learn new languages and adapt to different ecosystems.